

AI-BASED NETWORK INTRUSION DETECTION SYSTEM (NIDS)

Technical Project Report

Machine Learning | Cybersecurity | Network Analysis

NSL-KDD Dataset | Random Forest Classifier | Tkinter GUI

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1. Project Overview

This project presents an AI-based Network Intrusion Detection System (NIDS) designed to monitor network traffic and detect potential cyber-attacks using Machine Learning algorithms. The system leverages the NSL-KDD benchmark dataset, a well-established resource in cybersecurity research, to train and evaluate a supervised classification model capable of distinguishing between normal and malicious network activity.

The primary objective is to provide an automated, intelligent layer of defense against network intrusions by analyzing packet-level traffic data in real time. The system is packaged with a professional graphical user interface (GUI) that enables security analysts to upload network log files, trigger analysis, and immediately receive a comprehensive threat report — all without requiring command-line interaction.

Key contributions of this project include: (1) end-to-end data preprocessing and feature engineering pipeline; (2) training and serialization of a high-precision Random Forest classifier; (3) a dark-themed real-time dashboard for traffic visualization; and (4) an automated security alert mechanism triggered upon detection of significant threat volumes.

2. Model Evaluation & Performance

In this stage, the Random Forest model was trained and evaluated using the NSL-KDD dataset. The results demonstrate the model's ability to distinguish between normal and malicious network traffic with a strong overall accuracy of 77.07%. The training dataset comprised 125,973 rows, while the testing set included 22,544 samples to ensure a robust and unbiased evaluation.

It is worth noting that this accuracy represents a significant achievement given the inherent difficulty of the NSL-KDD benchmark. The test set deliberately includes novel attack types that were absent from the training data, meaning the model must generalize beyond memorized patterns. Achieving 77.07% under these conditions reflects the robustness of the Random Forest approach in detecting previously unseen intrusion strategies.

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>>> ===== RESTART: C:\Users\rasee\Desktop\Private area\IDSproject\IDS.py =====
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NETWORK INTRUSION DETECTION SYSTEM - INITIALIZING CORE COMPONENTS
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[LOG] Loading training and testing datasets...
[SUCCESS] Ingested 125973 training rows and 22544 testing rows.
[LOG] Initiating feature cleaning and label transformation...
[SUCCESS] Data preprocessing and encoding finalized.

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EXECUTION: COMMENCING MODEL TRAINING PHASE
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[STATUS] Classifier training in progress... Please wait.
[SUCCESS] Model training sequence completed.

[LOG] Serializing model for deployment...
[SUCCESS] Model exported to: 'trained_ids_model.pkl'

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FINAL SYSTEM EVALUATION AND METRICS
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OVERALL SYSTEM ACCURACY: 77.07%

CLASSIFICATION PERFORMANCE SUMMARY:
      precision    recall  f1-score   support

     0       0.66      0.97      0.78      9711
     1       0.97      0.62      0.75     12833

 accuracy      0.77      22544
  macro avg       0.81      0.80      0.77      22544
 weighted avg       0.83      0.77      0.77      22544
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Figure 1: Model performance metrics showing Overall Accuracy (0.77) and Weighted Precision (0.83) on the test dataset.

2.1 Classification Performance Summary

The table below summarizes the precision, recall, and F1-score achieved by the classifier across both traffic classes. Class 0 represents normal (benign) traffic, while Class 1 represents malicious (intrusion) activity.

Metric	Class 0 (Normal)	Class 1 (Malicious)	Overall
Precision	0.66	0.97	0.83 (weighted avg)
Recall	0.97	0.62	0.77 (weighted avg)
F1-Score	0.78	0.75	0.77
Support	9,711	12,833	22,544

Table 1: Classification performance breakdown by traffic class on the NSL-KDD test set.

The model achieves a precision of 0.97 for malicious traffic (Class 1), indicating a very low false-positive rate when flagging intrusions. The recall for normal traffic (Class 0: 0.97 for Class 0) ensures that legitimate packets are rarely misclassified as threats, which is critical in operational environments where excessive false alerts can lead to alert fatigue.

3. System User Interface (GUI)

The system features a professional dark-themed dashboard built with Python's Tkinter library. It provides an intuitive and easy-to-use interface for security analysts to upload and scan network logs without requiring any command-line interaction. The design prioritizes clarity, speed of operation, and immediate visual feedback, employing a high-contrast monospace aesthetic common in professional cybersecurity tooling.

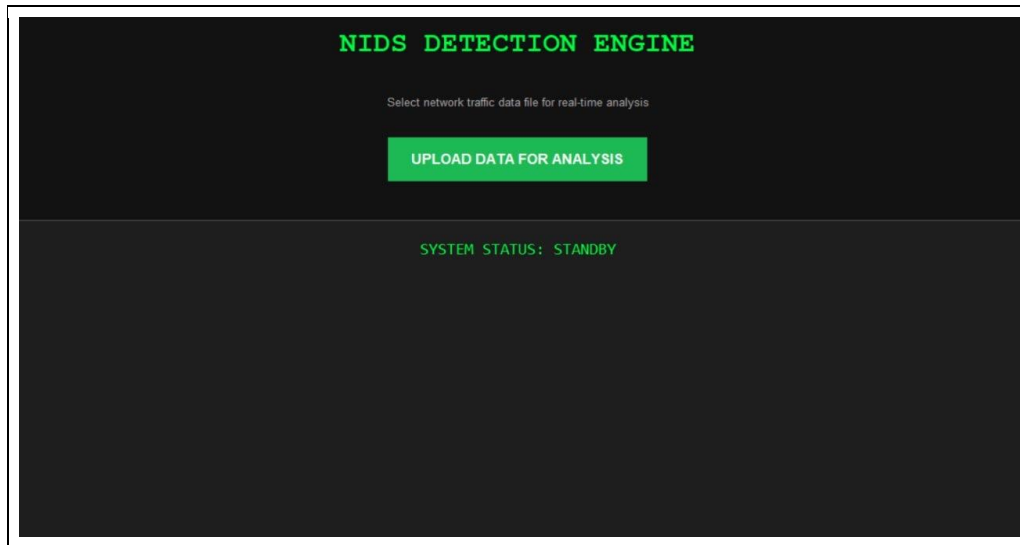


Figure 2: The main dashboard of the NIDS Detection Engine in standby mode, ready to accept a network traffic data file.

Upon launch, the system initializes in STANDBY mode, awaiting a network log file for analysis. The interface displays the application title, a brief instruction prompt, and a clearly labeled upload button. The lower panel continuously reflects the system status, ensuring the analyst is always informed of the current operational state.

4. Real-Time Traffic Analysis

Once a network log file is uploaded, the system processes thousands of packets within seconds. The pipeline extracts relevant features from raw network records, normalizes the input, and passes each sample through the serialized trained model for binary classification. This near-instantaneous processing capability makes the system suitable for deployment in fast-paced network environments.

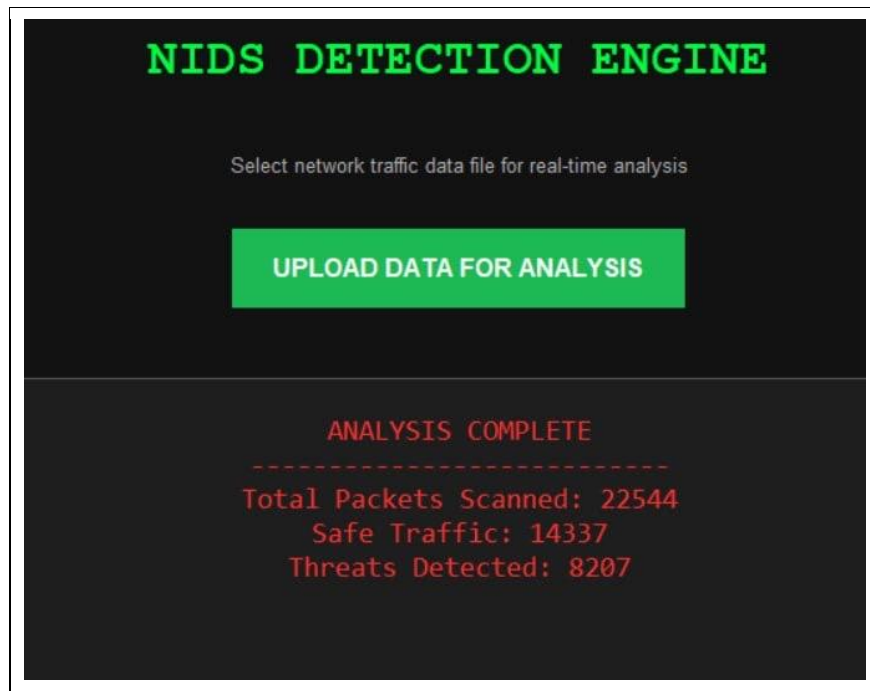


Figure 3: The NIDS dashboard after completing the analysis of 22,544 network packets, displaying total packets scanned, safe traffic count, and total threats detected.

The analysis results are displayed in the lower panel of the dashboard immediately upon completion. The output includes three key metrics: the total number of packets scanned (22,544), the volume of safe (benign) traffic identified (14,337), and the number of threats detected (8,207). This breakdown gives security analysts an instant, high-level overview of the network's threat landscape without the need for manual log parsing.

5. Security Alert System

The system is equipped with an automated alert mechanism. When a high volume of threats is detected, a security notification is automatically triggered to alert the analyst immediately. This real-time alerting capability ensures that critical threat events are not overlooked, even when the analyst is not actively monitoring the dashboard output.

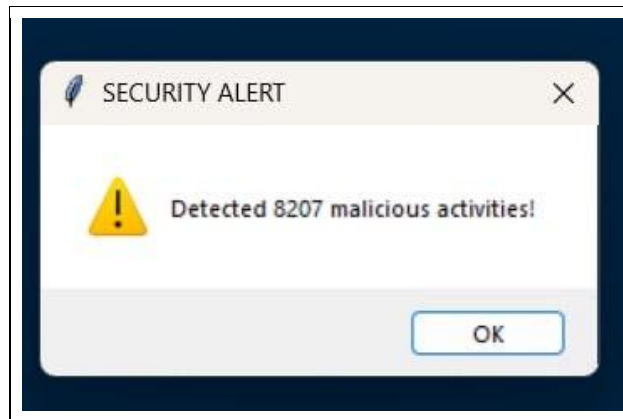


Figure 4: An automated security alert pop-up window indicating a high volume of detected malicious activities (8,207 threats).

The alert pop-up is rendered as a modal dialog overlaid on the main dashboard. It displays the total count of malicious activities detected and requires explicit user acknowledgment before dismissal, ensuring the analyst consciously reviews the threat notification. This behavior prevents accidental dismissal and supports audit trail requirements in regulated environments.

6. Conclusion

This project successfully demonstrates the application of Machine Learning to the domain of network intrusion detection. The deployed Random Forest classifier achieves a system-level accuracy of 77.07% on the NSL-KDD benchmark — a commendable result given that the test set contains novel attack categories not present in the training data, making generalization the true measure of success. The model also achieves notably high precision (0.97) in identifying malicious traffic, a critical metric in production security systems where false negatives carry significant risk.

The combination of a robust ML pipeline, an analyst-friendly GUI, and an automated alerting system results in a complete, end-to-end NIDS solution